Derick Freitas

dlfreitas@ucdavis.edu | (831)-529-6505 | dlfreitas.github.io

EDUCATION

University of California, Davis **Sept. 2024 - Present**

B.S. Mechanical Engineering *Expected Graduation: June 2028*

PROJECTS

Shift Buddy Device *C++ (Arduino, Visual Studio)*

- Led systems and interface design defining feature layout and device shift indicator logic for a manual transmission shifting assistant to teach novice manual drivers proper shifting timing
- Tuned RPM and throttle weights to model accurate shifting feedback under varying simulated driving conditions
- Integrated simulated vehicle sensor data from BeamNG.drive with an Arduino-based system using a C++ communication bridge built in Visual Studio

Ibanez TS9 Mounting Plate *FDM 3D Printing (Bambu Lab), SolidWorks*

- Iteratively prototyped a guitar pedal mounting plate to reinforce structural weaknesses identified through testing and customer feedback
- Improved mounting stability and strength by adding material supports around screw contacts and incorporated an interlocking support hook and dual alignment features

Fender Stratocaster CAD Model *SolidWorks*

- Modeled the neck, fretboard, headstock, frets, and inlays in a 4-person team to create a full-scale SolidWorks model of a Fender Stratocaster electric guitar using luthier multi-view drawings
- Produced part drawings and assembly in SolidWorks, mating to the body assembly for the complete model

EXPERIENCE

Sales Team Lead / Field Sales Representative **July 2025 – Oct. 2025**

Supreme Essence Inc. *Salinas, CA*

- Represented national brands such as AT&T within Costco locations across the Monterey Bay region
- Trained new hires on sales pitch, quoting, and use of sales and quoting software
- Averaged 30 new cellular lines per month generating over $7000 in commission
- Conducted interviews with prospective hires and contributed to team hiring decisions
- Served as a point of contact for team members with questions regarding customer needs and sales software

Freelance Jazz Guitarist **Sept. 2022 – Present**

Independent Contracting *Central California*

- Performed at venues, public events, and private wine events, working closely with event managers to deliver professional performances in various environments
- Directed setlists, rehearsal scheduling, musician communications, and sound setup logistics

SKILLS & COMPETENCIES

CAD & Design | SolidWorks (Modeling, Assemblies, Working Drawings), Fusion360 (Modeling, CAM),

Simulation & Analysis | MATLAB (Simulink, App Designer), C++ (Arduino, Visual Studio), Excel

Languages | Portuguese